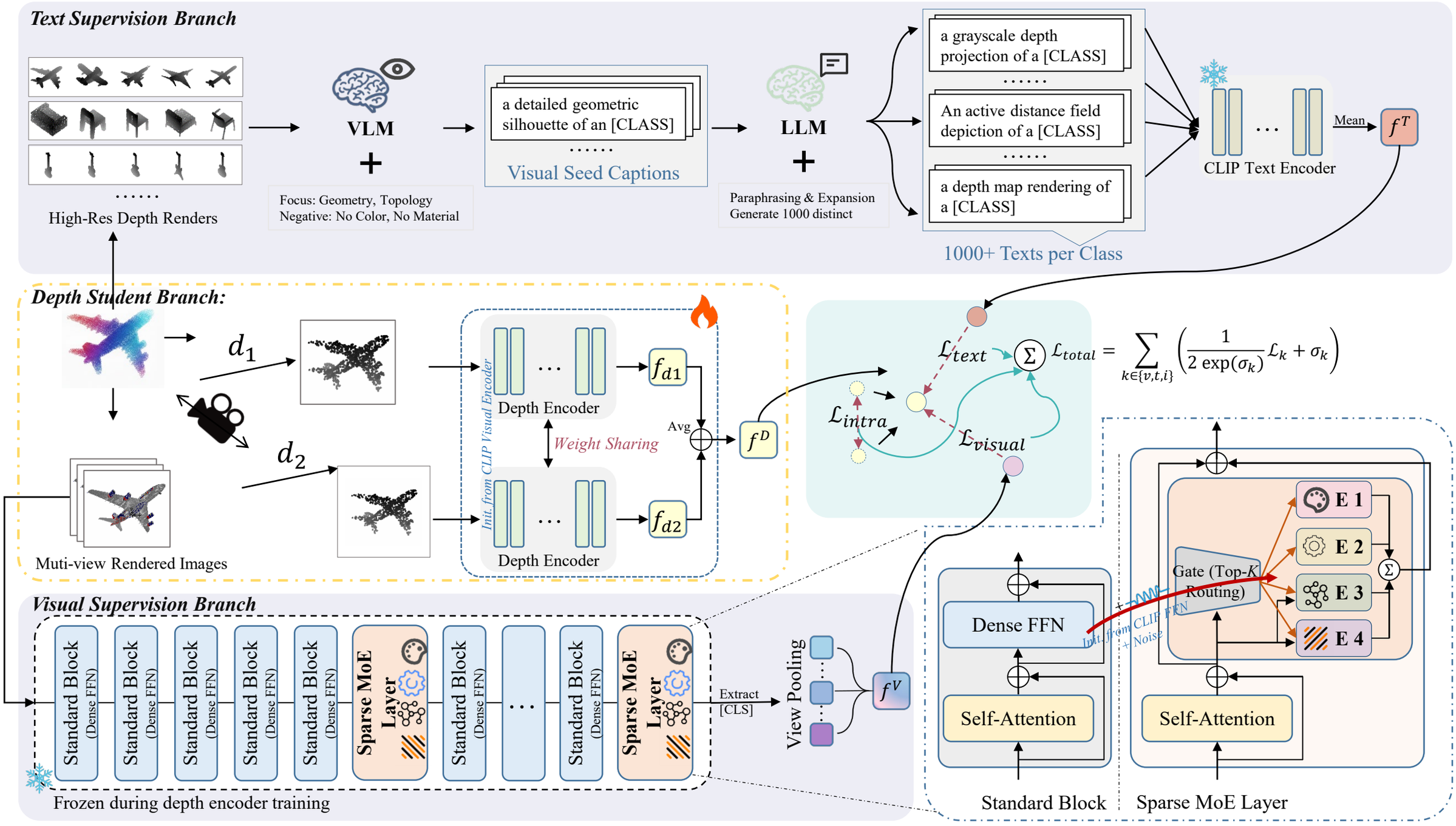
Text Supervision Branch
High-Res Depth Renders
VLM
+
Focus: Geometry, Topology
Negative: No Color, No Material
a detailed geometric silhouette of an [CLASS]
Visual Seed Captions
LLM
+
Paraphrasing & Expansion
Generate 1000 distinct
a grayscale depth projection of a [CLASS]
An active distance field depiction of a [CLASS]
a depth map rendering of a [CLASS]
1000+ Texts per Class
CLIP Text Encoder
Mean
f^T
Depth Student Branch:
d_1
d_2
Muti-view Rendered Images
Init. from CLIP Visual Encoder
Depth Encoder
Weight Sharing
Depth Encoder
f_{d1}
f_{d2}
Avg
f^D
$\mathcal{L}_{text}$
$\mathcal{L}_{intra}$
$\mathcal{L}_{visual}$
$\mathcal{L}_{total} = \sum_{k\in\{v,t,i\}} \left(\frac{1}{2\exp(\sigma_k)} \mathcal{L}_k + \sigma_k \right)$
Visual Supervision Branch
Standard Block (Dense FFN)
Sparse MoE Layer
Extract [CLS]
View Pooling
f^V
Frozen during depth encoder training
Dense FFN
Self-Attention
Standard Block
Gate (Top-K Routing)
E 1
E 2
E 3
E 4
Init. from CLIP FFN + Noise
Self-Attention
Sparse MoE Layer